SCIENCE & TECHNOLOGY

How much water do data centers use?

The answer has two parts: the water evaporated to cool the servers, and the water used to make their electricity.

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Plain metal cabinets beside a channel of still water. · AI-generated image

The digital world and artificial intelligence possess an invisible thirst. Data centers are massive electricity users; the IEA estimates they consumed about 415 terawatt-hours in 2024 (1.5 percent of global electricity), a figure expected to more than double by 2030 as AI grows. [1]

In many large facilities, the immense heat from this electricity is removed by evaporating water in cooling towers, and that evaporated water never returns to the pipe. When the water used by thermal power plants to generate that electricity is added, the toll grows heavier. Researchers estimated that training the GPT-3 model in Microsoft's American data centers could have evaporated about 700,000 liters of clean fresh water on site alone. [2] With every answer AI generates, it takes a sip of water from nature.

Sources

1. International Energy Agency – Energy and AI (2025)
https://www.iea.org/reports/energy-and-ai

2. Li, Yang, Islam & Ren – Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models
https://arxiv.org/abs/2304.03271